- All the non-zero digits are significant.
- All the zeros between two non-zero digits are significant, no matter where the decimal point is, if at all. → 2003 (4 significant digit)
- If the number is less than 1, the zero(s) on the right of decimal point but to the left of the first non-zero digit are not significant.

↳ 0.0002308 (4 signif.)

- The terminal or trailing zero(s) in a number without a decimal point are not significant.

↳ 12300 (3 signif.)

- The trailing zero(s) in a number with a decimal point are significant.

↳ 3.500 or 0.06900

(4 signif.)

2.02×10^3 → (3 signif.)

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(1.) Addition / Subtraction Rule

the final result should retain as many decimal places as are there in the number with the least decimal places.

$436.32 + 227.2 + 0.301 = 663.821$

$\therefore$ Ans $= 663.8$

(2.) Multiplication / Division

the final result should retain as many significant figures as are there in the original number with the least significant figures.

$2.73 \times 0.03 = 0.0819$

$2.73 \rightarrow 3$ $\quad$ $0.03 \rightarrow 1$

$\therefore$ Ans $= 0.08$

Rounding off:

1. $2.745 = 2.74$

EVEN

2. $2.735 = 2.74$

ODD

3. $2.5 = 2$

$3.5 = 4$

EDUNITI

Q 1. The area of a square is 5.29 cm^2. The area of 7 such squares taking into account the significant figures is (JEE MAIN 2019)

(a) 37 cm^2 (b) 37.030 cm^2

(c) 37.03 cm^2 (d) 37.0 cm^2

Soln: Add 5.29 seven time.

$5.29 + 5.29 + \cdots + 5.29 = 37.03$

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EDUNITI

Soln:

Q2. A student measuring the diameter of a pencil of circular cross-section with the help of a vernier scale records the following four readings 5.50 mm, 5.55 mm, 5.45 mm, 5.65 mm. The average of these four readings is 5.5375 mm and the standard deviation of the data is 0.07395 mm. The average diameter of the pencil should therefore be recorded as (2020)

(a) (5.5375 ± 0.0739) mm (b) (5.538 ± 0.074) mm

(c) (5.54 ± 0.07) mm (d) (5.5375 ± 0.0740) mm

Soln: After 2 places to Decimal.

$\therefore \quad 5.5375 \pm 0.07395$

$\Rightarrow 5.54 \pm 0.07$

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Soln:

Q3. For the four sets of three measured physical quantities as given below, which of the following options is correct? (2020)

(i) $A_1 = 24.36,\ B_1 = 0.0724,\ C_1 = 256.2$

(ii) $A_2 = 24.44,\ B_2 = 16.082,\ C_2 = 240.2$

(iii) $A_3 = 25.2,\ B_3 = 19.2812,\ C_3 = 236.183$

(iv) $A_4 = 25,\ B_4 = 236.191,\ C_4 = 19.5$

(a) $A_1 + B_1 + C_1 < A_2 + B_2 + C_2 = A_3 + B_3 + C_3 < A_4 + B_4 + C_4$

(b) $A_1 + B_1 + C_1 = A_2 + B_2 + C_2 = A_3 + B_3 + C_3 = A_4 + B_4 + C_4$

(c) $A_1 + B_1 + C_1 < A_3 + B_3 + C_3 < A_2 + B_2 + C_2 < A_4 + B_4 + C_4$

(d) $A_4 + B_4 + C_4 < A_1 + B_1 + C_1 < A_3 + B_3 + C_3 < A_2 + B_2 + C_2$

$A_1 + B_1 + C_1 = 280.6324$
$= 280.6$

$A_2 + B_2 + C_2 = 280.722$
$= 280.7$

$A_3 + B_3 + C_3 = 280.6642$
$= 280.7$

$A_4 + B_4 + C_4 = 280.691$
$= 281$

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